

Aidan Liu

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<https://arti-dev.github.io/>

Education

University of Virginia, School of Engineering and Applied Science

Expected May 2027

BS, Major: Computer Science, Minor: Data Science, GPA: 3.863

Organizations: First Year Players, Student Game Developers, UVA Dependable Systems and Analytics Lab

Relevant Coursework: Data Structures and Algorithms II, Intro to Cybersecurity, Computer Systems and Organization II, Software Engineering, Defense Against the Dark Arts

Experience & Volunteering

Robotic Telesurgery Undergraduate Researcher, UVA-DSA Group

July 2026 – present

- Migrated an older UDP-based framework used for research surrounding telesurgery topics into a modular ROS2 architecture, improving system component interchangeability and maintainability.
- Engineered tooling for an autonomous recovery system to assist surgeons when network conditions degrade.

Motion Planning Software Engineer, Cavalier Autonomous Racing

May 2025 – July 2026

- Joined the highly selective team following its **2024 Indy Autonomous Challenge victory**
- Collaborated with the team to develop *cavsim*, an environment to research, test, and evaluate planning algorithms for our autonomous racecar, accelerating iteration cycles and reducing real-world testing risk.
- Traveled to Laguna Seca to support on-site testing at the Indy Autonomous Challenge in Summer 2025.

Event Staff Student Manager, University of Virginia Event Management

January 2025 - present

- Leading student staff and coordinating event logistics to support thousands of campus events each year.
- Facilitating day-to-day operations of the University's student hub through clear communication, problem-solving, and peer support.

Projects

Waste Classification Machine Learning Project

Python, PyTorch, Jupyter

Trained and evaluated convolutional neural networks (CNNs) on a 20,000-image dataset to classify waste as organic or recyclable. Performed data preprocessing, exploratory analysis, and data augmentation to improve model generalization and accuracy, achieving 94% accuracy with a transfer-learned CNN.

Carpool Buddy Rideshare Platform for University Organizations

Django, Python, Heroku

Built a full-stack web application designed for university clubs to organize rideshares for events, including user profiles, messaging, and moderation tools. Served as the go-to development lead for a five-person team to debug complex issues, resolve Git merge conflicts, and support development and deployment using Django and Heroku.

Honors and Certifications

International Collegiate Programming Competition, Regional Competition

2023-2025

Consistent top 10 placements out of ~90 teams across three years in team algorithmic programming competitions.

Virginia Science Olympiad, State Competition

2022-2023

Multiple top 5 placements including 1st place in Flight in 2023.

First Tee Greater Washington, DC: Jonathan Wildes Memorial Scholarship for \$2,500

2023

Tripoli Rocketry Association: Level 1 High Power Certification

2024

Skills

Programming Languages/Tools: Python, Java, HTML/CSS, GDScript, IntelliJ, WSL, GitHub Actions, SQL, C++

Languages: English, Chinese (Intermediate)